

# Spectromorphology

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*Composing with Algorithms*  
<http://www.bjarni-gunnarsson.net>

# Spectromorphology

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*“In the confusing, wide-open sound-world, composers need criteria for selecting sound materials and understanding structural relationships. So descriptive and conceptual tools which classify and relate sounds and structures can be valuable compositional aids.”*

**Denis Smalley**

***Spectromorphology*** is a framework/set of tools for understanding structural behaviours as experienced during the listening process.

***Spectromorphology*** is not a compositional theory but can influence compositional methods.

# Spectromorphology

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Smalley has published **three key texts** on the subject.

*"Spectro-morphology and Structuring Processes" (1986)*

*"Listening Imagination: Listening in the Electroacoustic Era" (1992)*

*"Spectromorphology: Explaining Sound Shapes" (1997).*

“Sonic behavior has been linked to causality, but it is also associated with relationships that exist among sounds that have been placed in a context. Smalley introduces related terms, including dominance (foreground)/subordination (background) and conflict/coexistence, to illustrate other behavioral aspects of sound.”

**(Leigh Landy)**

# Spectromorphology

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An attempt is made to ignore the sound-producing technology where **technological listening** (when “a listener ‘perceives’ the technology or technique behind the music rather than the music itself”) is thought to bring clichés whereas **Spectromorphological** thinking focuses on the perceived music.

Intended for **acousmatic music** that focuses on spectral qualities instead of notes, on motion rather than metrical time.

- \* Focus on sounds as abstract objects
- \* Emphasis on timbre, texture, and gesture
- \* De-emphasis of traditional pitch and rhythm

# Spectromorphology

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*“The word 'spectromorphology' therefore refers literally to the way sound spectra (spectro-) are shaped (-morphology) over time. It may be defined as 'an approach to sound materials and musical structures which concentrates on the spectrum of available pitches and their shaping in time'. [...] Denis Smalley introduces a concept of correspondence to describe moments of shift from one morphology to another. He describes an attack which: sets in motion a resonating spectrum which we might expect to decay gradually. The further away we travel from the attack point the more we may be coaxed into observing how the spectral components of the continuant phase proceed, and the less interested we shall be in the generic role of the initial impact. This example is typical of spectromorphological thinking, and Smalley goes on to elaborate how it might form a structuring principle for composition.”*

**(The Digital Musician)**

# Reduced Listening

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**Source bonding**, *"the natural tendency to relate sounds to supposed sources and causes, and to relate sounds to each other because they appear to have shared or associated origins."*

**Reduced listening (écoute réduite)** is an abstract, concentrated listening to sound events where spectromorphological properties are uncovered. It requires the distractions of source bonding to be blocked.

# Reduced Listening

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All sounds have both ***concrete*** and ***abstract*** potential for interpretation.

***Reduced listening*** concentrates on the abstract properties of a sound, its amplitude and spectral qualities and how these evolve in time.

# Listening as a Composer

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*“Many composers regard reduced listening as an ultimate mode of perceptual contemplation. But it is as dangerous as it is useful for two reasons. Firstly, once one has discovered an aural interest in the more detailed spectromorphological features, it becomes very difficult to restore the extrinsic threads to their rightful place. Secondly, microscopic perceptual scanning tends to highlight less pertinent, low-level, intrinsic detail such that the composer–listener can easily focus too much on background at the expense of foreground.”*

**(Dennis Smalley)**

# Morphology

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The temporal shaping of a sound is often referred to as its *morphology*.

In the case of acoustical sounds, increased energy produces increased spectral richness. So does the process when a sound arrives toward a listener providing a better detail when perceiving it. This also influences the dynamic evolution a sound's amplitude in time.

Morphologies can be used to categorize and describe sounds. In spectromorphology this feature is also discussed in terms of being *functional* and *generative* for musical development.

# Gestures and Textures

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Smalley defines two main categories of sound, gestures and textures.

- \* ***Gesture*** as a specific motion to or from a goal. Involves energy and consequences, intervention, growth and causality.

- \* ***Texture*** as concerned with internal behavior patterning, it is continuing, self-propagating and implies non-processive time.

# Gesture

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*“The notion of gesture as a forming principle is concerned with propelling time forwards, with moving away from one goal towards the next goal in the structure – the energy of motion expressed through spectral and morphological change. Gestural music, then, is governed by a sense of forward motion, of linearity, of narrativity. The energy– motion trajectory of gesture is therefore not only the history of an individual event, but can also be an approach to the psychology of time.”*

**(Denis Smalley)**

# Gesture

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*“Sound-making gesture is concerned with human, physical activity which has spectromorphological consequences: a chain of activity links a cause to a source [...] A gesture is therefore an energy–motion trajectory which excites the sounding body, creating spectromorphological life.”*

*“A knowledge of sounding gesture is culturally very strongly imbedded. [...] It is particularly important for acousmatic music where the sources and causes of sound-making become remote or detached from known, directly experienced physical gesture and sounding sources. The process of increasing remoteness I refer to as gestural surrogacy.”*

**(Denis Smalley)**

# Gestural Surrogacy

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**First-order surrogacy** “projects the primal level into sound, and is concerned with sonic object use in work and play prior to any ‘instrumentalisation’”

**Second-order surrogacy** “is traditional instrumental gesture, a stage removed from the first order, where recognisable performance skill has been used to develop an extensive registral articulatory play.”

**Third-order surrogacy** “is where a gesture is inferred or imagined in the music. The nature of the spectromorphology makes us unsure about the reality of either the source or the cause, or both.”

**Remote surrogacy** “is concerned with gestural vestiges. Source and cause become unknown and unknowable as any human action behind the sound disappears.”

# Gestural Surrogacy

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*"I venture to suggest that an electroacoustic music which is confined to the second order does not really explore the potential of the medium, while a music which does not take some account of the cultural imbedding of gesture will appear to most listeners a very cold, difficult, even sterile music."*

**(Dennis Smalley)**

# Texture

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*“Texture is concerned with internal behaviour patterning, energy directed inwards or reinjected, self-propagating; once instigated it is seemingly left to its own devices; instead of being provoked to act it merely continues behaving.”*

**(Denis Smalley)**

# Archetypes

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Three features are common to most sounds, how they start, continue and stop. This is named as ***onset***, ***continuant*** and ***termination*** by Smalley.

He identifies three basic envelope shapes of instrumental archetypes:

## 1 ***Attack-impulse Archetype***

(Sudden onset and immediate decay)

## 2 ***Attack-decay Archetype***

- Closed (attack, brief resonance, decay)
- Open (attack, sustain, decay)

## 3 ***Graduated Continuant Archetype***

- Gradual behaviour, focused on continuation, rather than attack

# Spectromorphological Expectation

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Three important temporal phases: **onset**, **continuant** and **termination**. There are also three important spectromorphological archetypes found in instrumental usage:

- \* The **attack** alone. This is a momentary energetic impulse.
- \* The **attack-decay**. The attack is extended by a resonance
- \* The **graduated continuant**. Where all three phases are present.

The archetypes train thinking of **spectromorphological expectation**.

The required understanding of spectral change "*provides an almost 'natural' reference-base not only for developing the wider, more imaginative spectromorphological repertory into the third-order surrogacy of electroacoustic music, but for decoding patterns of expectation in musical form.*"

# Structural Functions

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## **onsets**

departure

emergence

anacrusis

attack

upbeat

downbeat

## **continnants**

passage

transition

prolongation

maintenance

statement

## **terminations**

arrival

disappearance

closure

release

resolution

plane

(Concerned with expectation patterns)

# Structural Functions

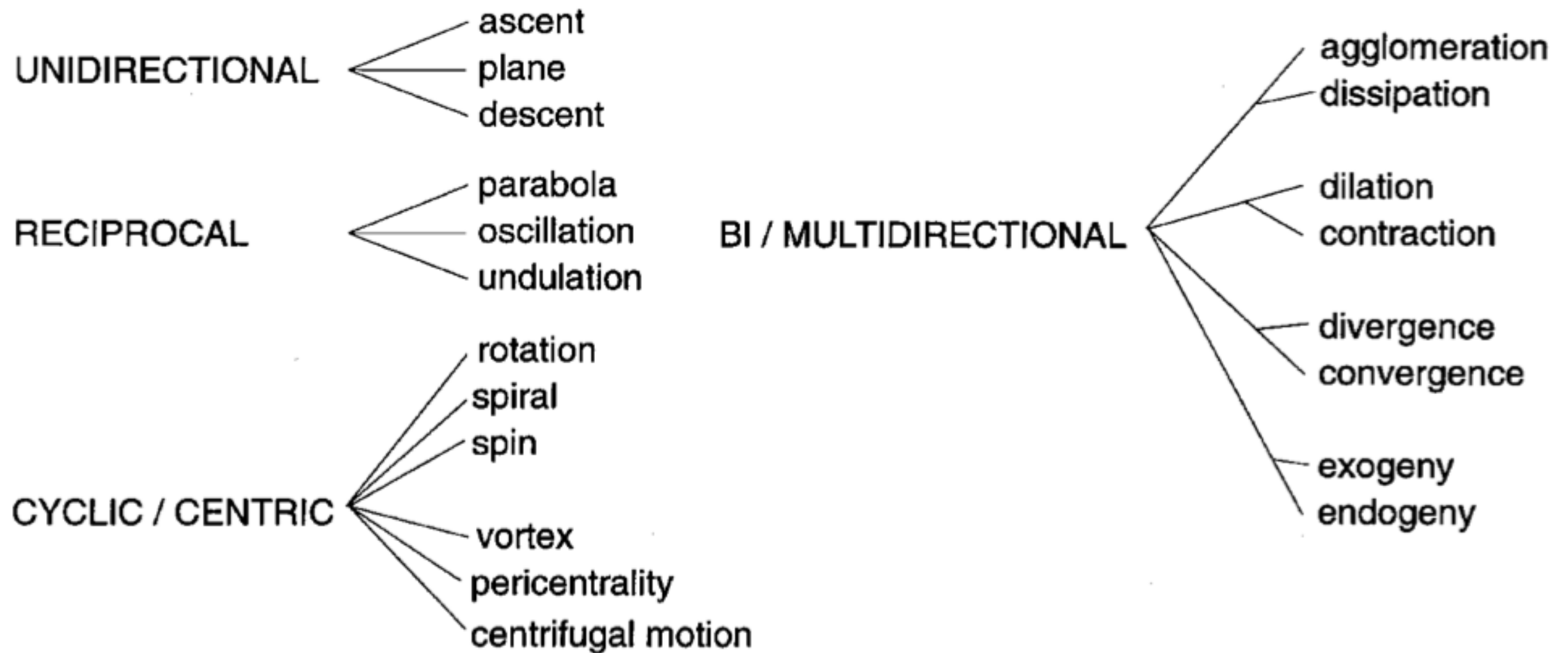
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The attribution of which function a particular event has several important considerations.

1. Function attribution is *not normally a conscious* thought-process.
2. Function attribution is a *continuing, incomplete process*, subject to revision.
3. Function attribution may be *double or ambiguous*.
4. There is *no clear temporal border* between the three function-types.

# Motion and Growth

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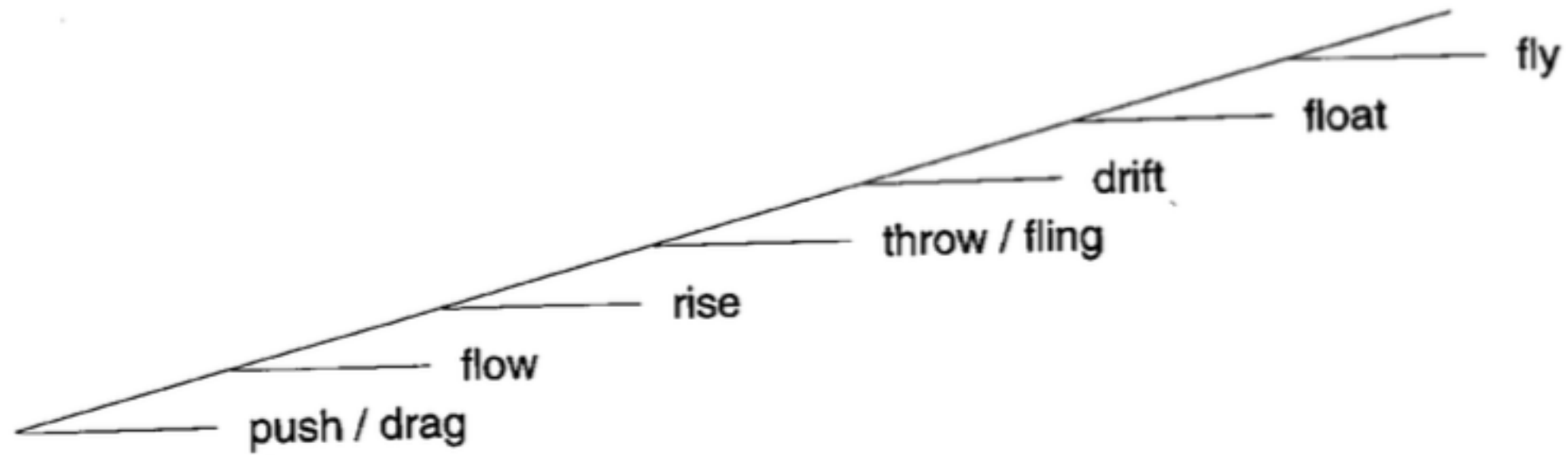


**Figure 1.** Motion and growth processes.

(Concerned with expectation patterns)

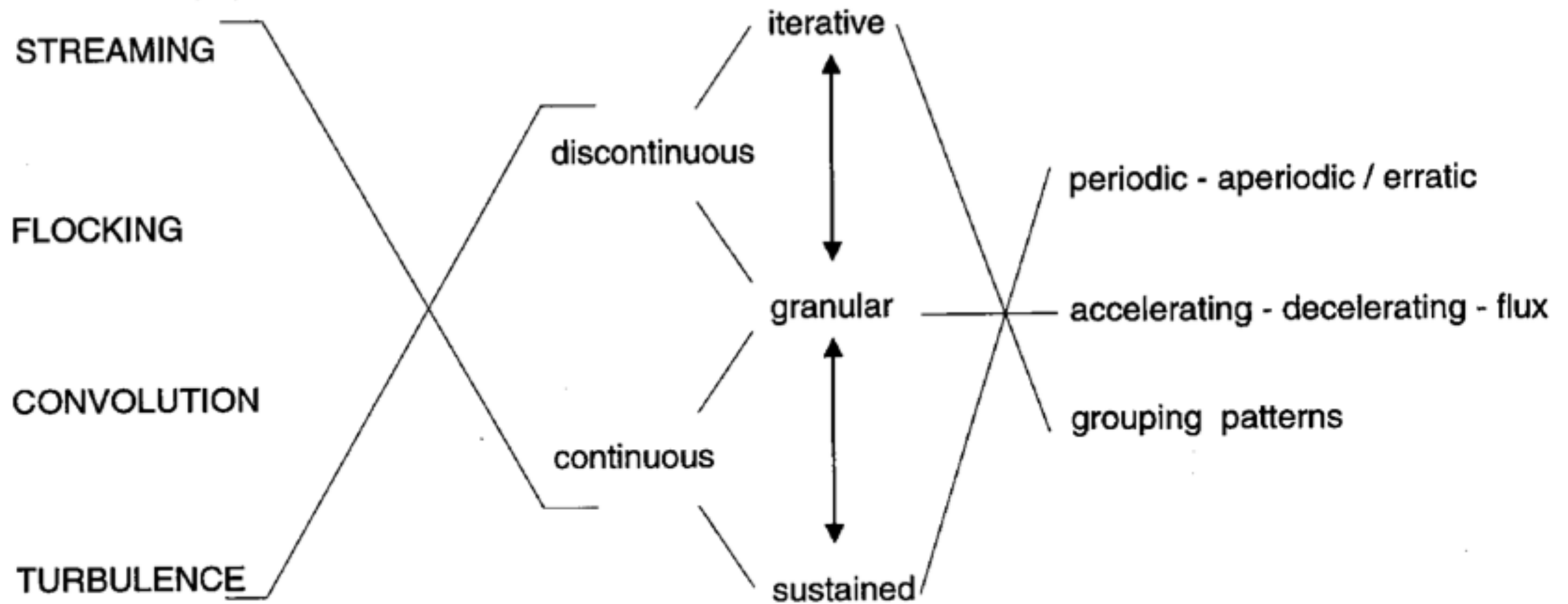
# Characteristic Motions

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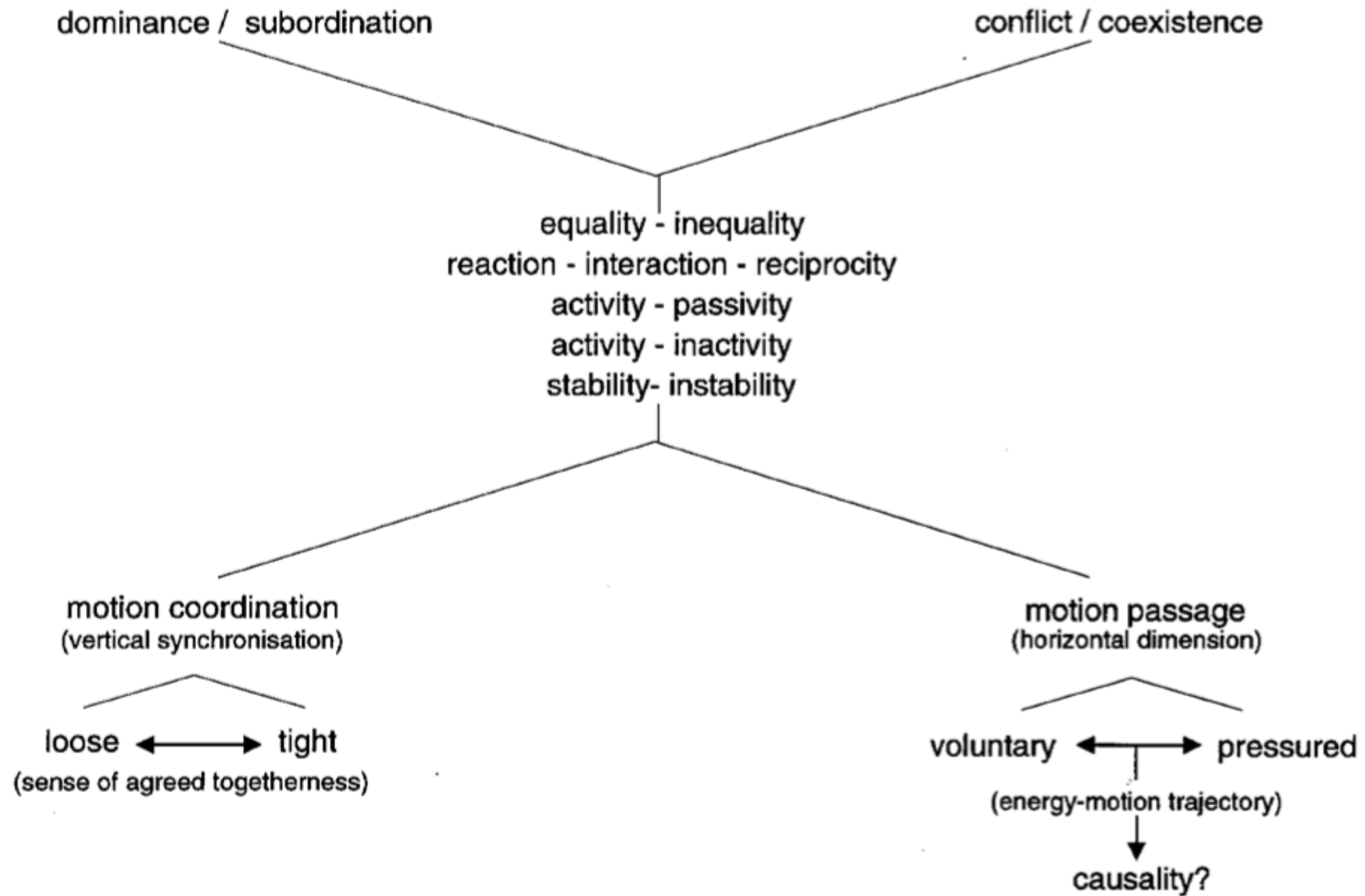
(Seven characteristic motions.)

# Texture Motion



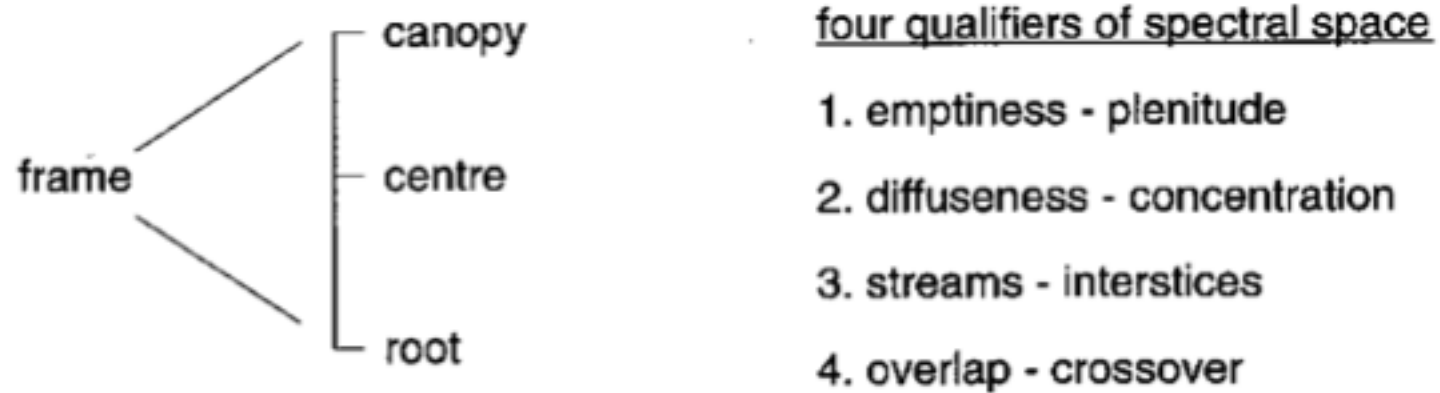
**Figure 3.** Texture motion.

# Behavior

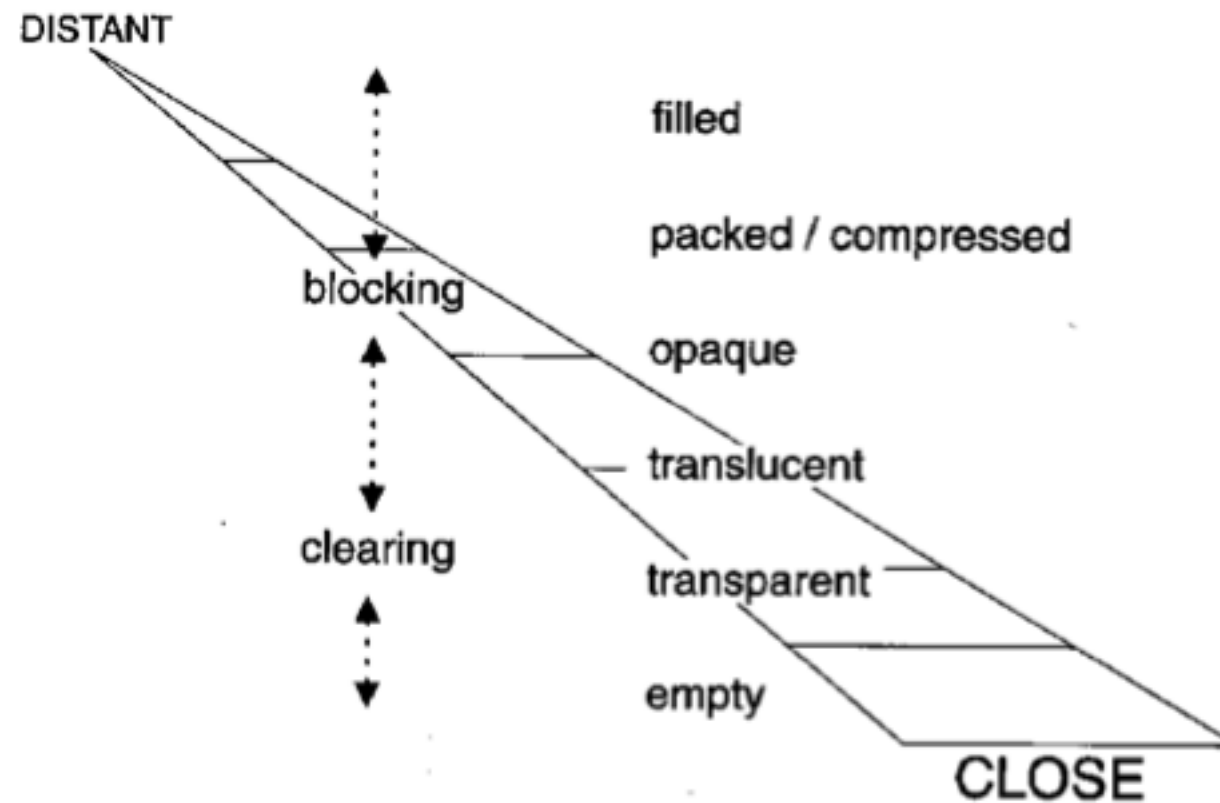


**Figure 4.** Behaviour.

# Spectral Space

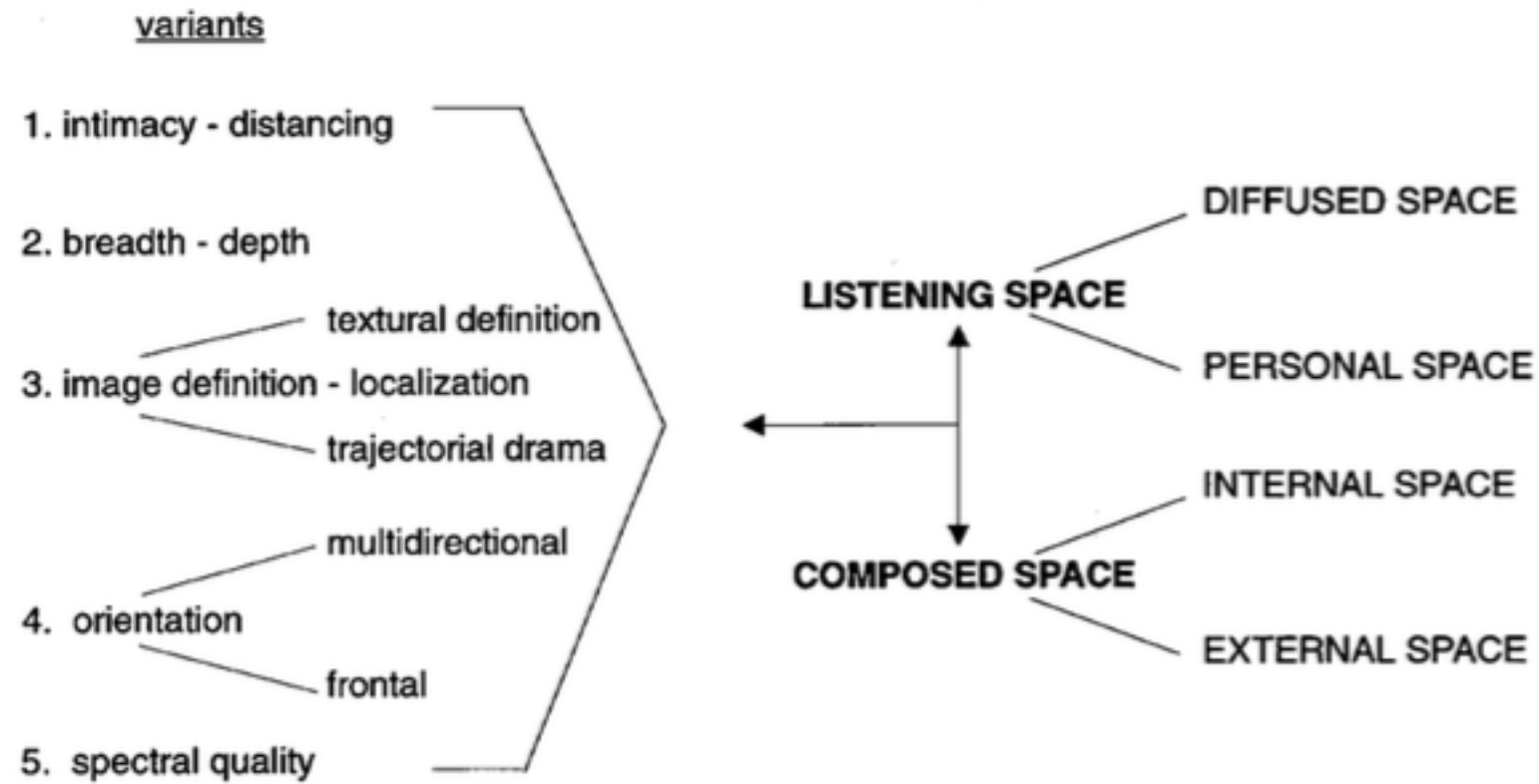


**Figure 6.** Occupancy of spectral space.



**Figure 7.** Spectral density.

# Composed and Listening Spaces



**Figure 8.** Composed and listening spaces.

# External Space

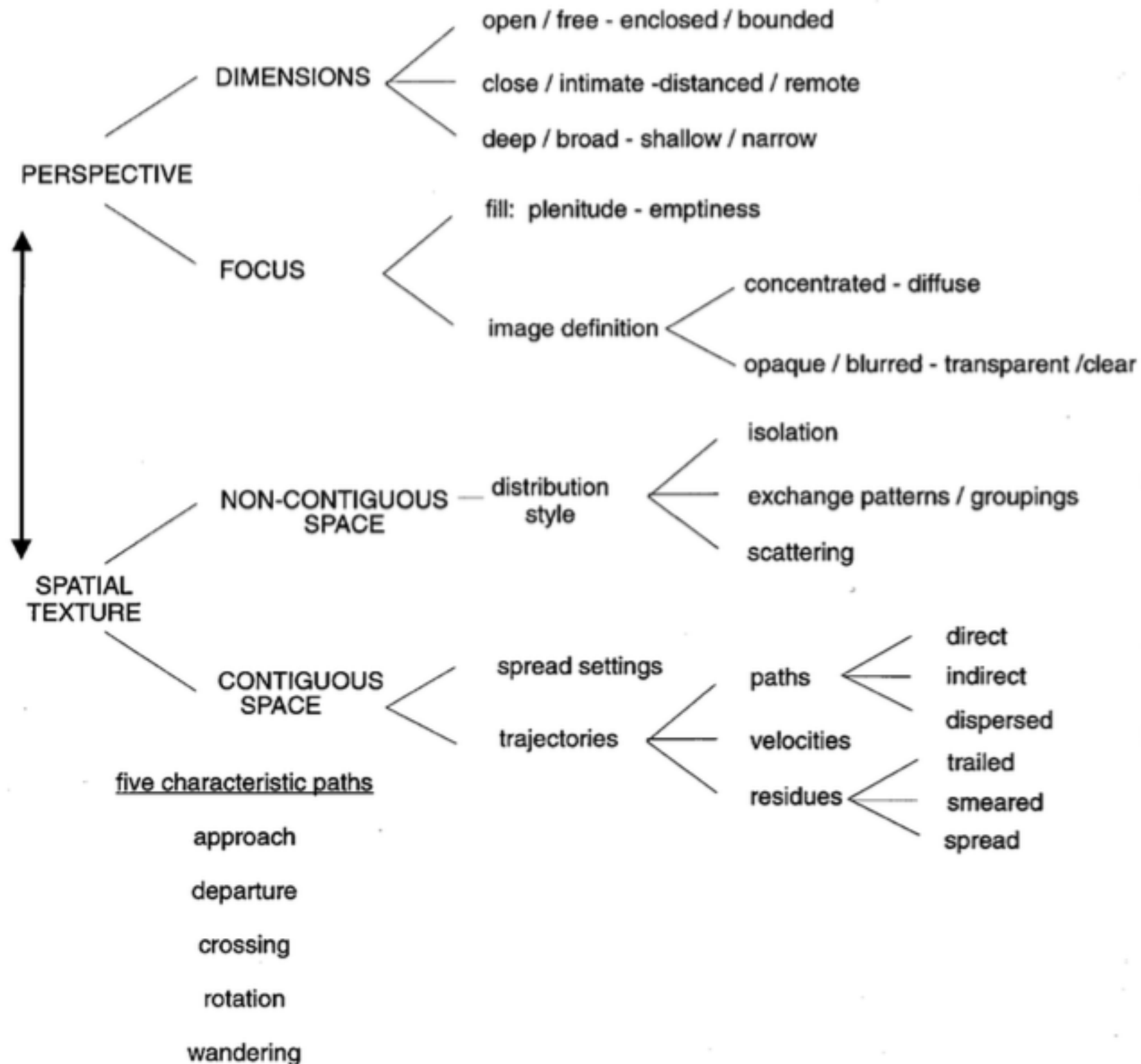


Figure 9. External space.

Denis Smalley - Valley Flow (1992)



# Valley Flow

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*“The formal shaping and sound content of Valley Flow were influenced by the dramatic vistas of the Blow Valley in the Canadian Rockies. The work is founded on a basic flowing motion which is stretched or contracted to create floating and flying contours, panoramic sweeps, and the flinging out of textural materials. [..]The result is a kind of sonic landscape which seems to link directly with a visual experience. The piece is layered and striated, just like the landscape it depicts, but also sweeps in a speeded-up version of the activity that formed the Blow Valley itself or the passage of the eye across its various features. The detailed manipulation of the sound through time, however, also creates a psychological effect that seems to correspond to the sense of awe experienced in the presence of such large geological formations.”*

**(The Digital Musician)**

# Composition Possibilities

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**Material choice.** Using the word *emergence*, sounds can be chosen on the basis of becoming more apparent over time, while when using *disappearance*, sounds that fulfilled this criteria are those that faded out over time. This process can inform the selection of sounds to be recorded and choose from within a given sound library.

**Sound sculpting.** Vocabulary can also inform *shaping* and *manipulation* of single sounds. Choosing attack, passage and release can dictate the sculpting of a single sound through the accentuation of onset abruptness, (*attack*), duration extension (*passage*), and emphasis upon a 'letting go' of tension (*release*).  
sound-unit creation.

# Composition Possibilities

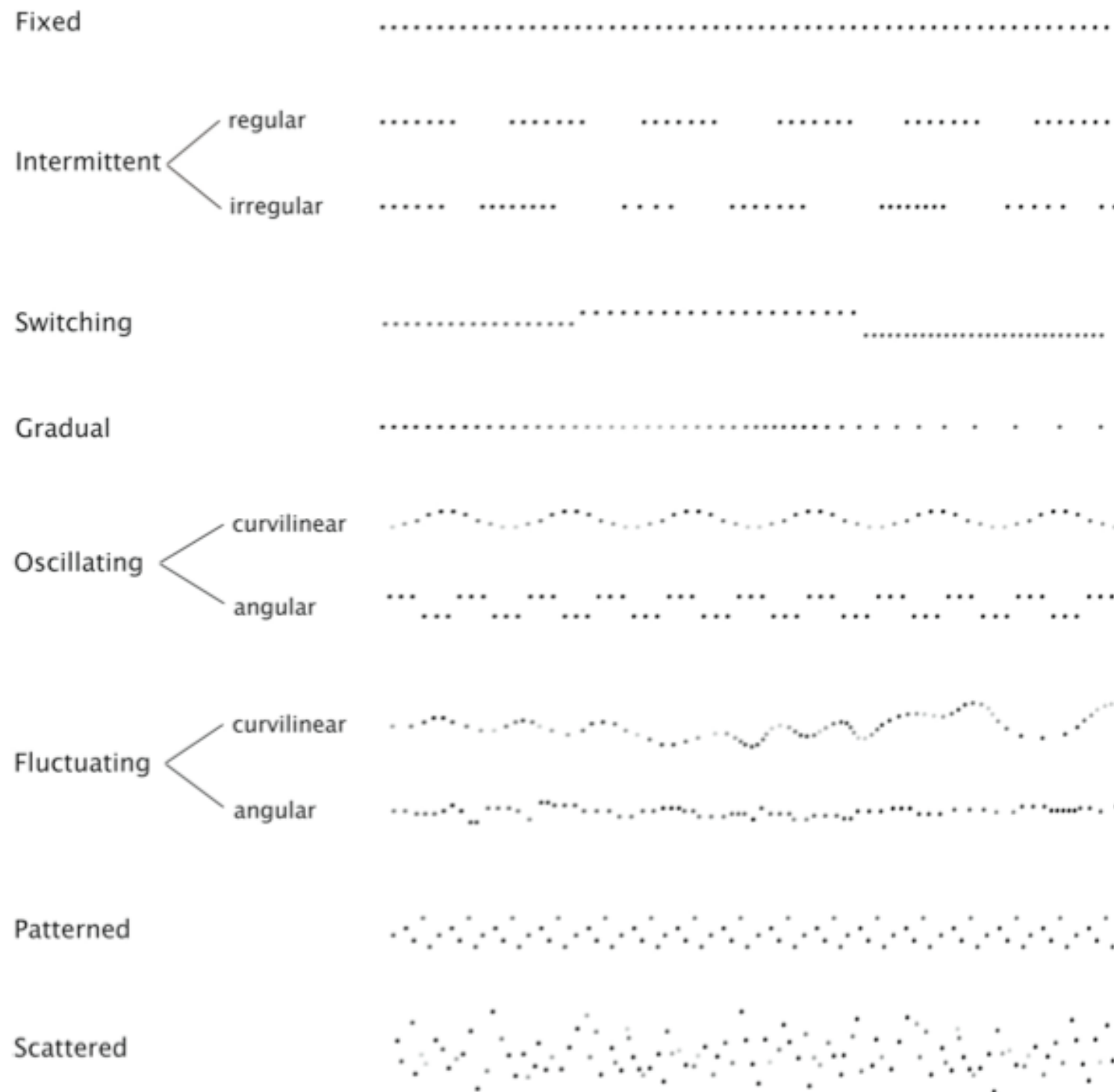
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## *Structure creation.*

Attack, passage and release can also be applied to three separate sounds assembled together to form a new shape, a 'sound unit'. Sound units can be strung together to form longer phrase lengths called morphological strings.

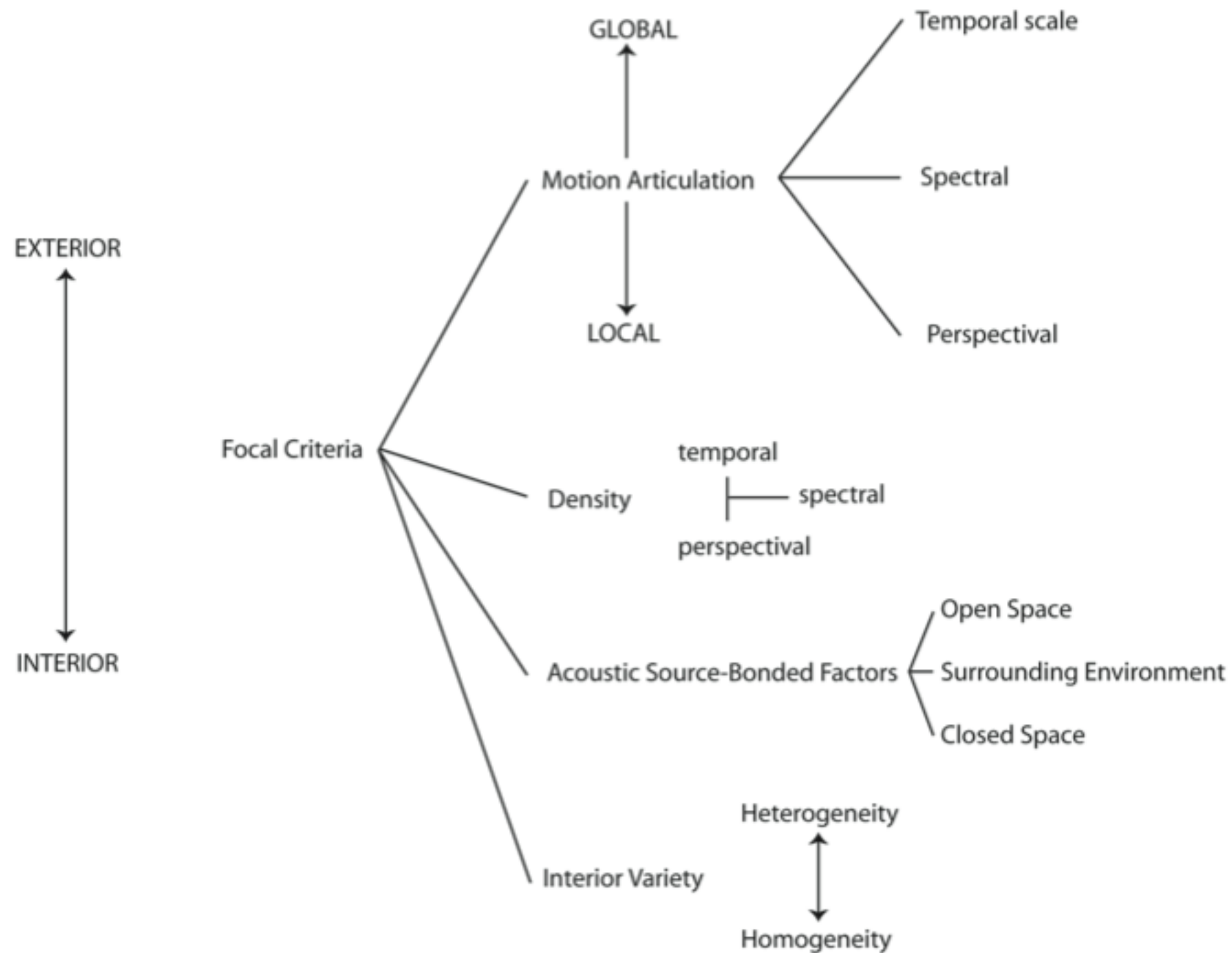
*(Manuella Blackburn)*

# Textons



**Figure 3.** Texton propagation modes.

# Motion Focal Schemas



**Figure 10:** Motion focal schemas.

# Motion Directionality

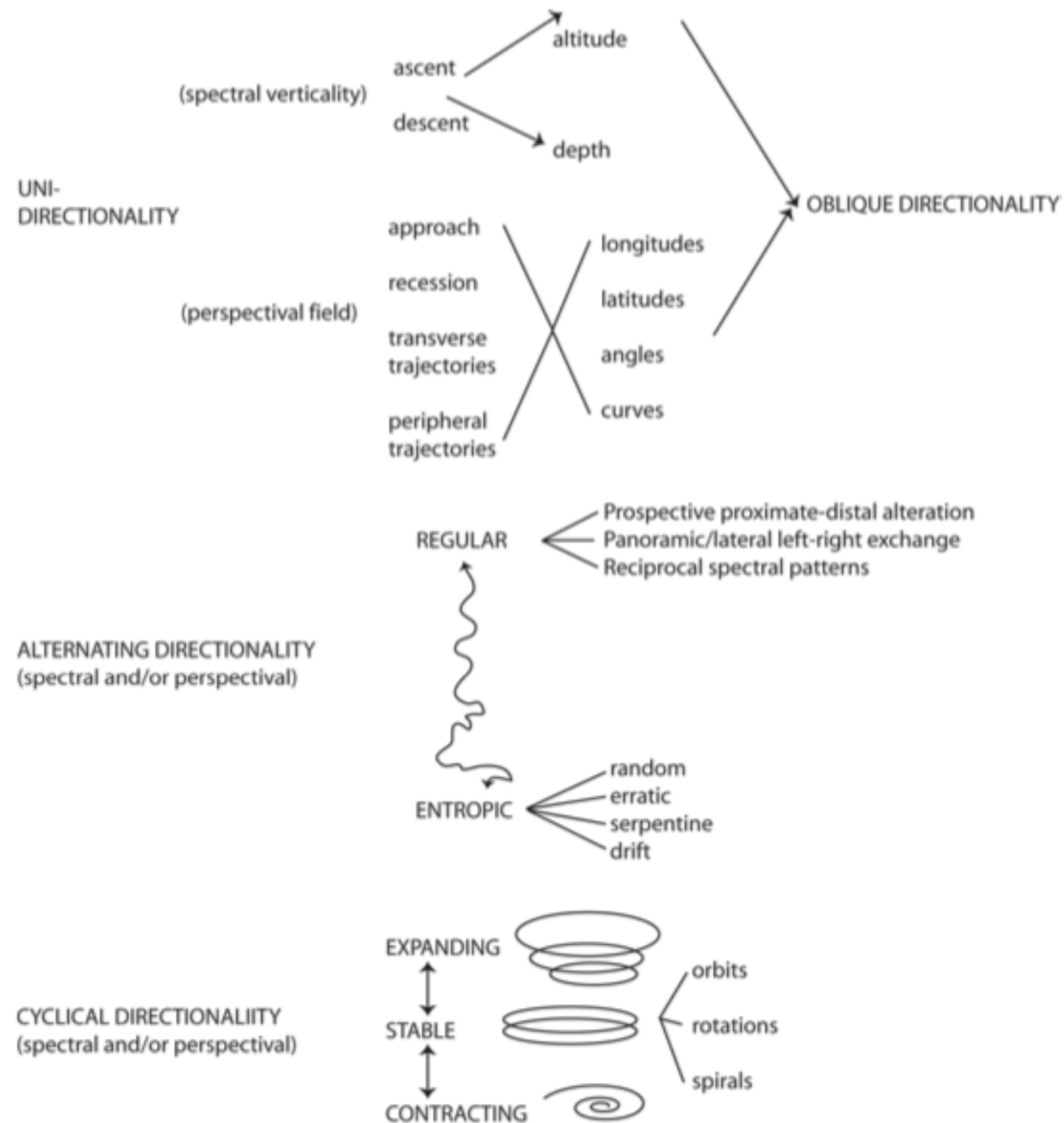
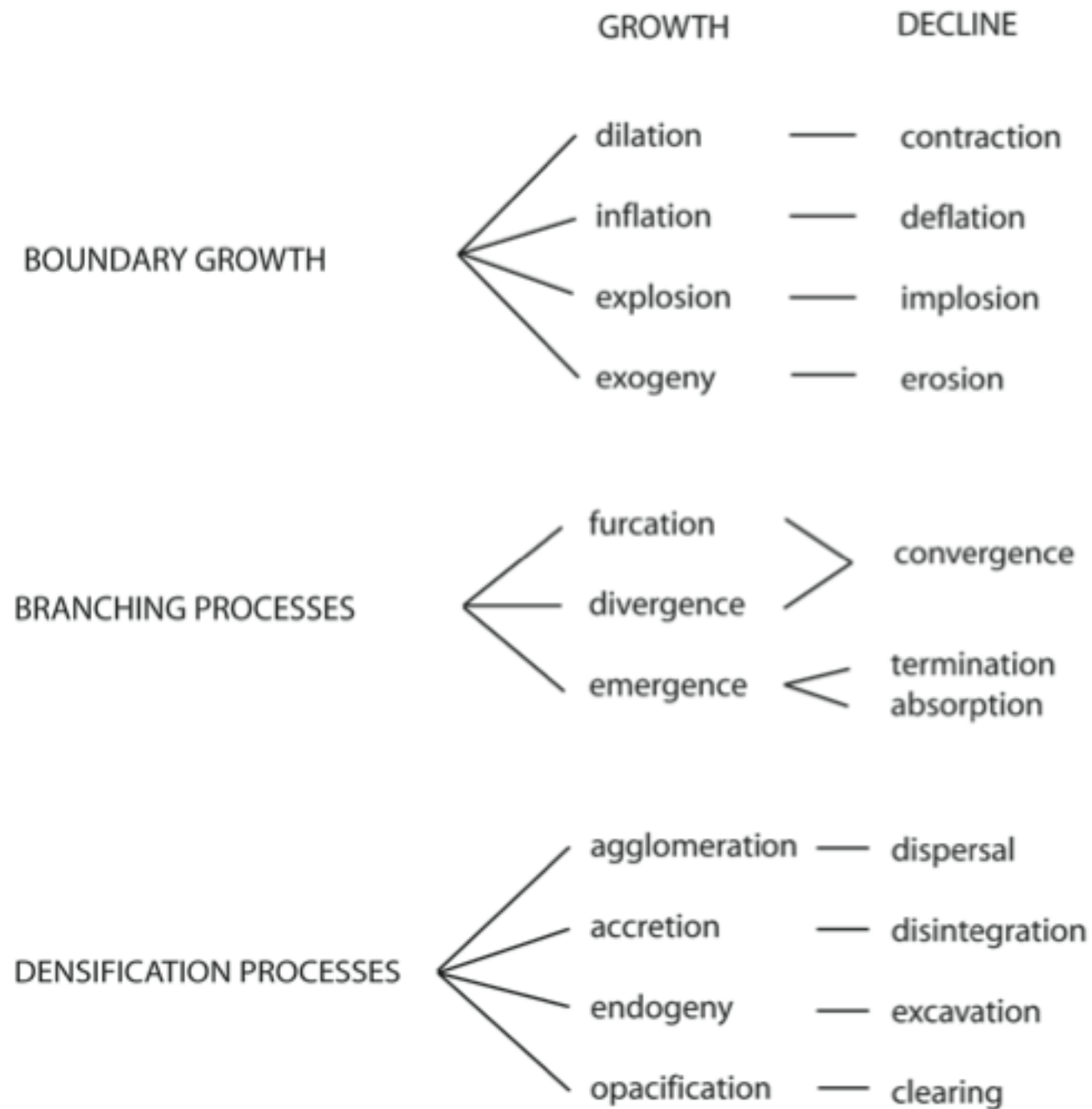


Figure 11: Motion directionality.

# Growth Processes

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**Figure 12:** Growth processes.

Erik Nyström  
Far-from-equilibrium (2008)



# Exercises

# Exercises - CDP

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1. Choose a short sound for input, **extend** it (using drunk, zigzag or other extend methods) and finally implement a time varying (using breakpoints) **distortion** based on wavecycles where the perceived distortion increases in time.
2. Implement a **morphing** between two sounds where the contact points of the material should flow smoothly together. In order to obtain a successful morphing, a potential first step could be to transform them in similar ways before executing the morphing.
3. Implement an **algorithmic transformation** where four sounds are extended in four different ways.

# Exercises - Spectromorphology

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1. Compose the ***overlapping*** of two textures. The first one should contain clear ***iterative motion*** with a ***simple directionality*** while the the second one remains more ***constant*** and ***concerned*** with ***local, internal*** activity.
2. Compose a ***streaming texture*** exposing ***granular behavior*** with a slightly accelerating motion. The texture could consist of synthetic or sample-based particles.
3. Compose ***four layers*** of sound where each occupies a special role in the overall ***spectral space***. A role might mean a certain layer is ***filled, empty*** or ***diffused*** in relation to other layers. These sound-layers may interact and vary in time.

